Comp-4400: Axiomatic Semantics (3 points)

Last Name	First Name	ID	Email ID	Signature	Marks	Marked by
				Lucas Serrut		

1 (2 points) Write the weakest precondition for the following Hoare triples.



Weakest Pre-condition

A predicate P is weaker than a predicate Q if $Q \Rightarrow P$. Given a program and a post-condition, there can be many pre-conditions. For example, `false` is a pre-condition for any program and post-condition. Hence, we are only interested in the weakest pre-condition. When you write the weakest pre-condition, also make sure that you simplify it as far as possible. For example, when you have $x = 5 \wedge true$, it should be simplified as $x = 5$. If you have $x = 5 \wedge x = 3$, you should simplify it into $false$.

1.
$\{\quad\}$
$x := x + k;$
$\{x = 7 \wedge k = 4\}$

2.
$\{\quad\}$
$x := x + 1;$
$x := x - 2;$
$\{x = 5\}$

3.
$\{\quad\}$
$z := x;$
$x := y;$
$y := x;$
$\{x = 5 \wedge y = 3\}$

4.
$\{\quad\}$
$x := y;$
$z := x;$
$y := x;$
$\{x = 5 \wedge y = 3\}$

5.
$\{\quad\}$
$z := x;$
$x := y;$
$y := z;$
$\{x = 5 \wedge y = 3\}$

6.
$\{\quad\}$
$x := y;$
$\{x > y\}$

7.
$\{\quad\}$
$x := y;$
$\{x \geq y\}$

8.
$\{\quad\}$
if $(x > y)$ then $x := y;$
$\{x = y\}$

9.
$\{\quad\}$
if $(x > y)$ then $x := y;$
$\{x \geq y\}$

10.
$\{\quad\}$
if $(x > y)$ then $x := y;$
$\{x \leq y\}$

2 (1 point) Write the loop invariants for the following programs:

1.
```
{true}
f := 1; n:=10;
while (n \= 0) do ( f := n*f; n:=n-1; )
{f=10!}
```

The loop invariant is:

2.
```
{true}
f := 1; n:=10;
while (n > 0) do ( f := n*f; n:=n-1; )
{f=10!}
```

The loop invariant is:
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